

```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 45 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a=1;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(r==1||c==1||r==nr||c==nc)printf("%3c",'*');
else printf("%3d",a++);
}
printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

2:36 PM
18-Dec-24

```
TC
Enter no of rows and columns 10 10
* * * * *
* 1 2 3 4 5 6 7 8 *
* 9 10 11 12 13 14 15 16 *
* 17 18 19 20 21 22 23 24 *
* 25 26 27 28 29 30 31 32 *
* 33 34 35 36 37 38 39 40 *
* 41 42 43 44 45 46 47 48 *
* 49 50 51 52 53 54 55 56 *
* 57 58 59 60 61 62 63 64 *
* * * * *
```

`int a=1;`

`if(r==1 || c==1 || r==nr || c==nc) p(*);
else p(a++);`

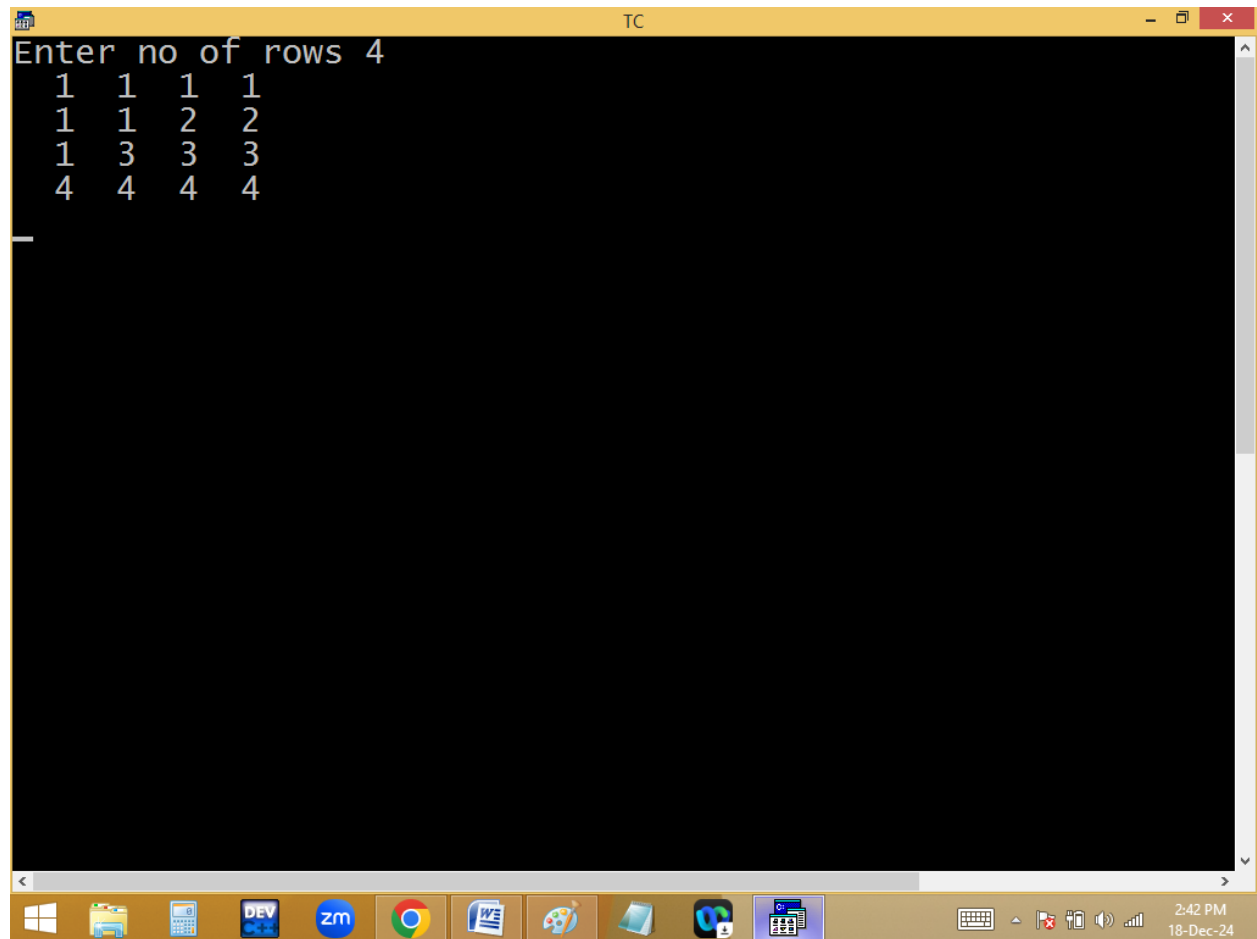
```
*****
*1234*
*5678*
*****
```

```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 20 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows " );
scanf("%d",&n);
for(r=1 ;r<=n;r++)
{
for(c=1;c<=n;c++)
{
if(c<=n-r)printf("%3d",1);
else printf("%3d",r);
}
printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

2:42 PM
18-Dec-24

```
TC
Enter no of rows 4
1 1 1 1
1 1 2 2
1 3 3 3
4 4 4 4
```



```

Enter no of rows 10
1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 2 2
1 1 1 1 1 1 1 3 3 3
1 1 1 1 1 1 4 4 4 4
1 1 1 1 1 5 5 5 5 5
1 1 1 1 6 6 6 6 6 6
1 1 1 7 7 7 7 7 7 7
1 1 8 8 8 8 8 8 8 8
1 9 9 9 9 9 9 9 9 9
10 10 10 10 10 10 10 10 10 10

```

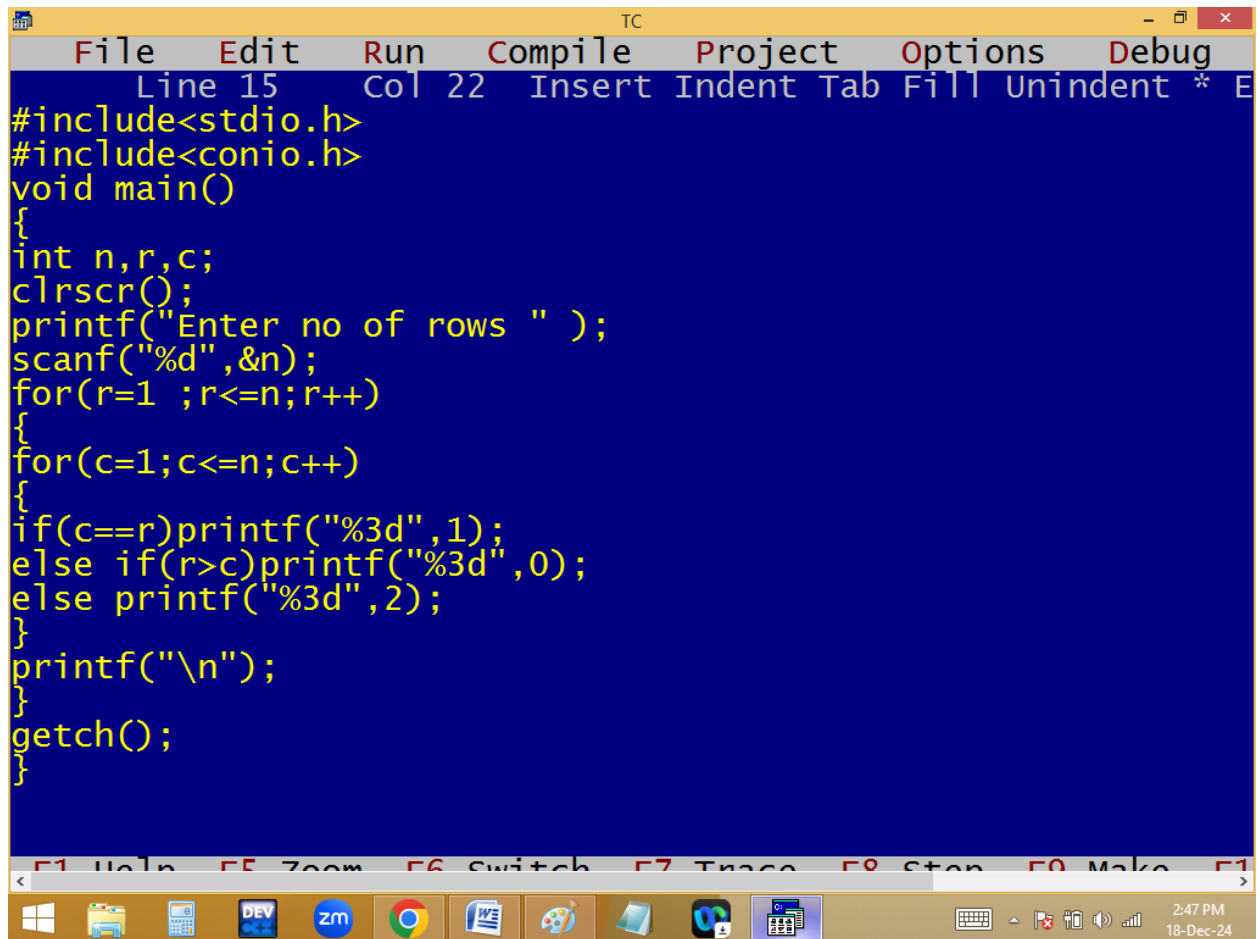
```

for(r=1;r<=4;r++)
{
  for(c=1;c<=4;c++)
  {
    if(c<=n-r)p(1);else p(r);
  }
  p("\n");
}

```

$$\begin{array}{rcl}
 n & & r \\
 4 & - & 1 = 3 \\
 4 & - & 2 = 2 \\
 4 & - & 3 = 1 \\
 4 & - & 4 = 0
 \end{array}$$

1	1	1	1
1	1	2	2
1	3	3	3
4	4	4	4

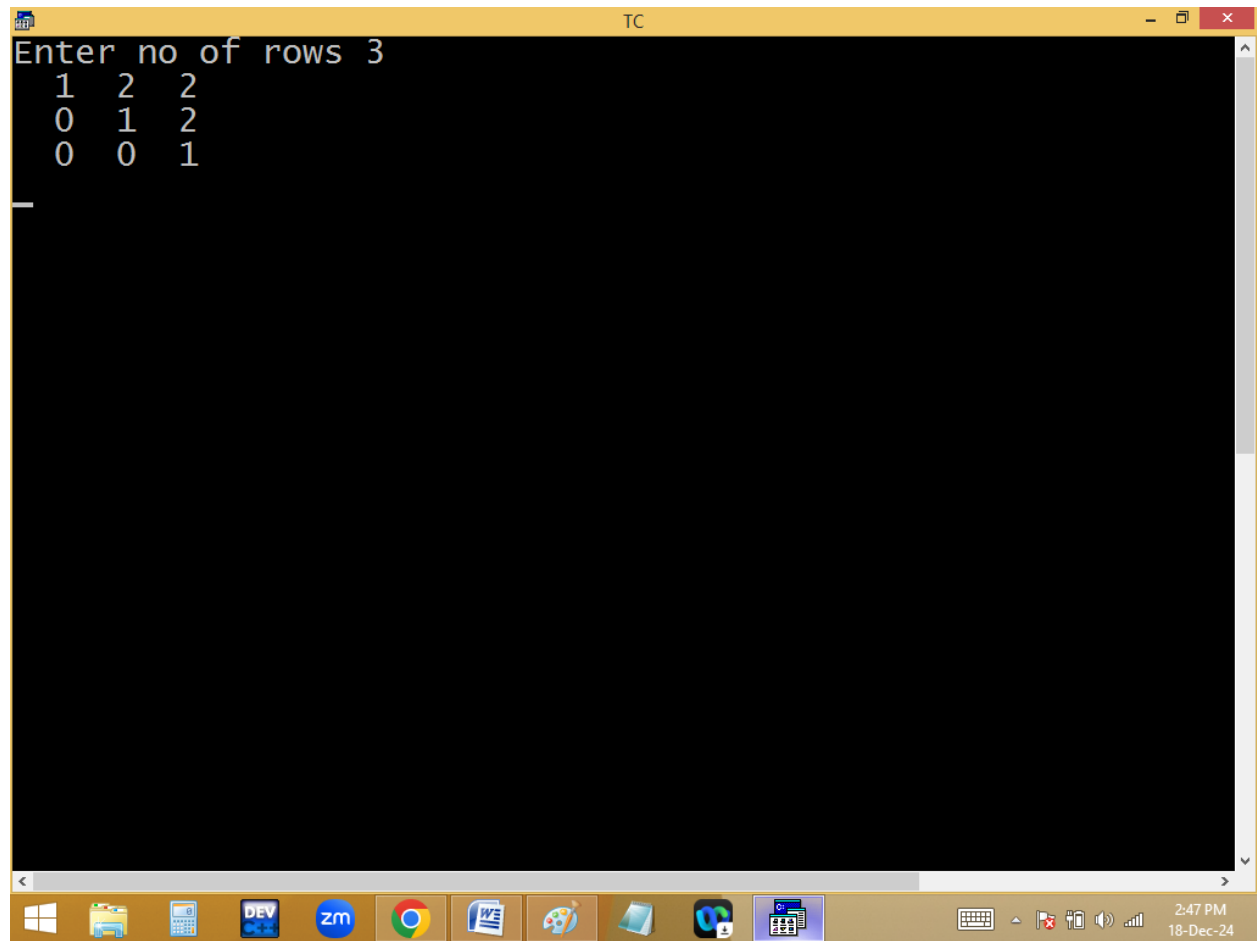


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar indicates "Line 15" and "Col 22". The main editing area has a blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows " );
scanf("%d",&n);
for(r=1 ;r<=n;r++)
{
for(c=1;c<=n;c++)
{
if(c==r)printf("%3d",1);
else if(r>c)printf("%3d",0);
else printf("%3d",2);
}
printf("\n");
}
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions. Below the toolbar, a status bar shows the time "2:47 PM" and the date "18-Dec-24".

```
TC
Enter no of rows 3
1 2 2
0 1 2
0 0 1
```



The image shows a Windows 10 desktop environment. A terminal window titled "TC" is open, displaying the prompt "Enter no of rows 3" and a 3x3 matrix of numbers: 1 2 2, 0 1 2, and 0 0 1. The taskbar at the bottom features icons for Windows, File Explorer, Calculator, DEV, Zoom, Chrome, Word, Paint, Edge, and a calendar. The system clock in the bottom right corner indicates the time is 2:47 PM on 18-Dec-24.

```
TC
Enter no of rows 10
1 2 2 2 2 2 2 2 2 2
0 1 2 2 2 2 2 2 2 2
0 0 1 2 2 2 2 2 2 2
0 0 0 1 2 2 2 2 2 2
0 0 0 0 1 2 2 2 2 2
0 0 0 0 0 1 2 2 2 2
0 0 0 0 0 0 1 2 2 2
0 0 0 0 0 0 0 1 2 2
0 0 0 0 0 0 0 0 1 2
0 0 0 0 0 0 0 0 0 1
```



```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 43 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows " );
scanf("%d",&n);
for(r=1 ;r<=n;r++)
{
for(c=1;c<=n;c++)
{
if(c==r){textcolor(LIGHTRED);cprintf("%3d",1);}
else if(r>c){textcolor(LIGHTGREEN);cprintf("%3d",0);}
else {textcolor(YELLOW);cprintf("%3d",2);}
}
printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

2:49 PM
18-Dec-24

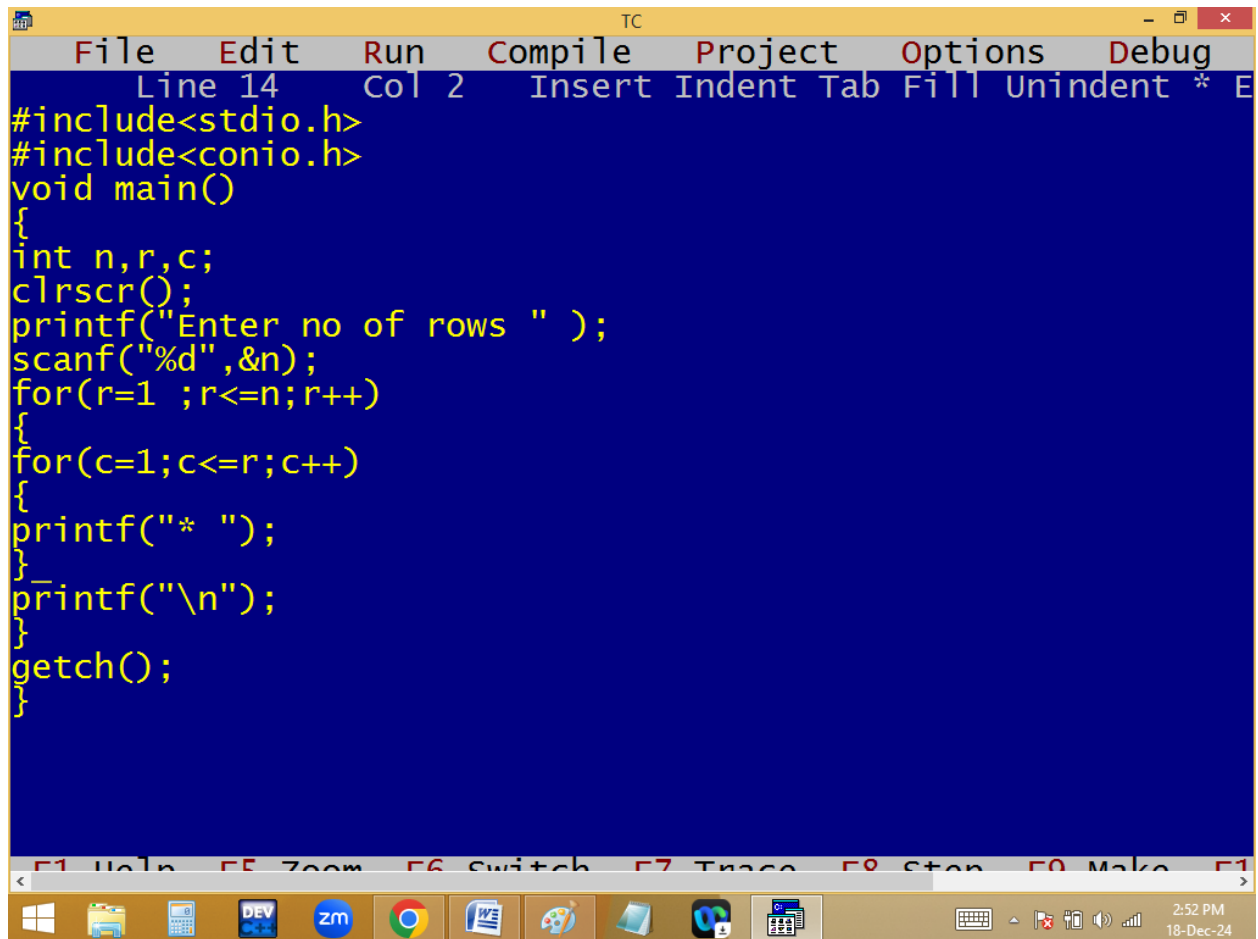
```
TC
Enter no of rows 9
1 2 2 2 2 2 2 2 2
0 1 2 2 2 2 2 2 2
0 0 1 2 2 2 2 2 2
0 0 0 1 2 2 2 2 2
0 0 0 0 1 2 2 2 2
0 0 0 0 0 1 2 2 2
0 0 0 0 0 0 1 2 2
0 0 0 0 0 0 0 1 2
0 0 0 0 0 0 0 0 1
```

```
if( r==c ) p(1);
else if(r>c) p(0);
else p(2);
```

1	2	2
1,1	1,2	1,3

0	1	2
2,1	2,2	2,3

0	0	1
3,1	3,2	3,3



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 14", "Col 2", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code in yellow text:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows " );
scanf("%d",&n);
for(r=1 ;r<=n;r++)
{
for(c=1;c<=r;c++)
{
printf("* ");
}
printf("\n");
}
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a folder. The system tray on the right shows the time as 2:52 PM and the date as 18-Dec-24.

```
TC
Enter no of rows 10
*
* *
* * *
* * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
for( r=1; r<=4; r++ )
{
    for( c=1; c<=r; c++) p(*);
    p("\n");
}
```

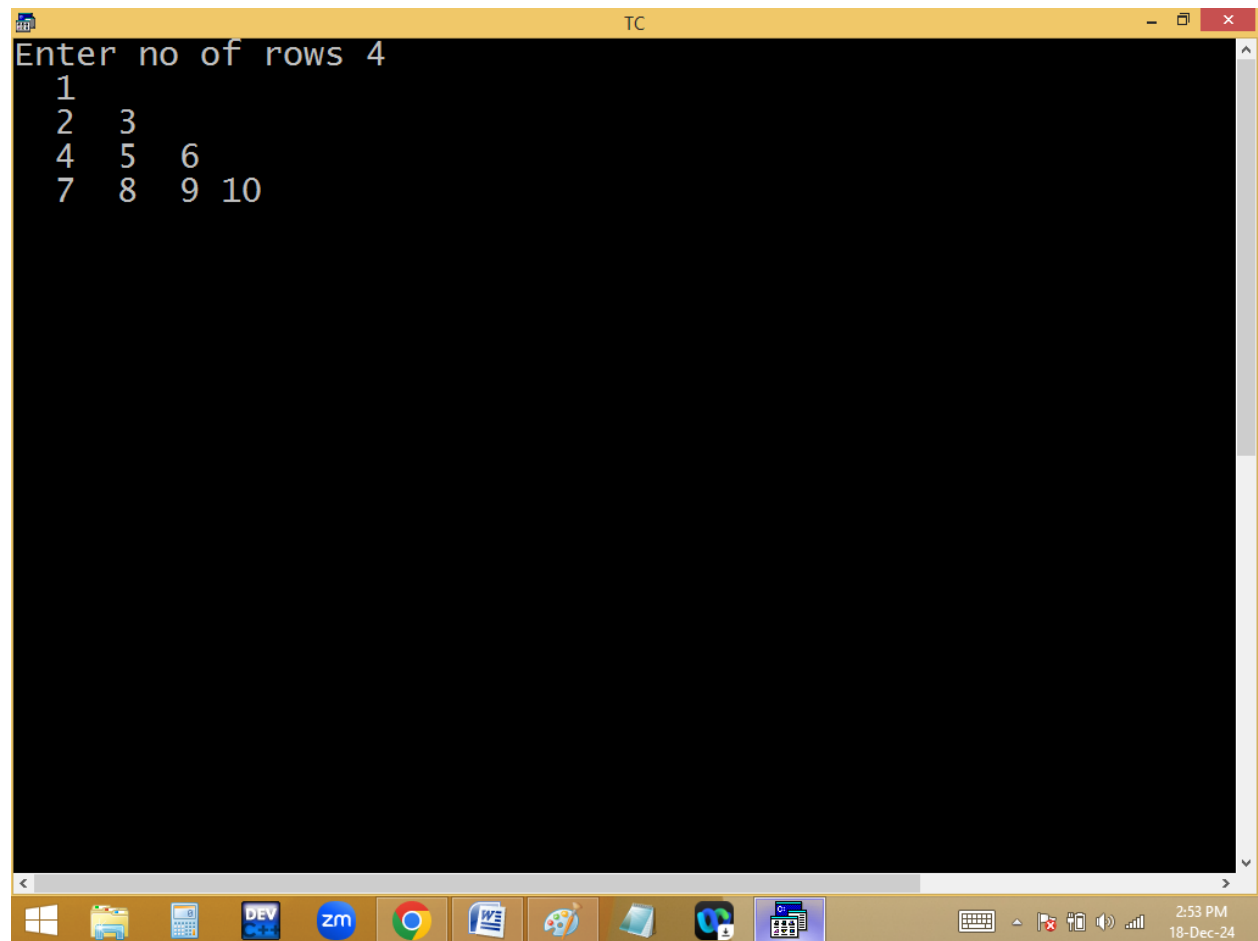
<u>c = 1 to</u>	<u>r</u>		
1	1	—	* — 1
1 to 2	2	—	** — 2
1 to 3	3	—	*** — 3
1 to 4	4	—	**** — 4

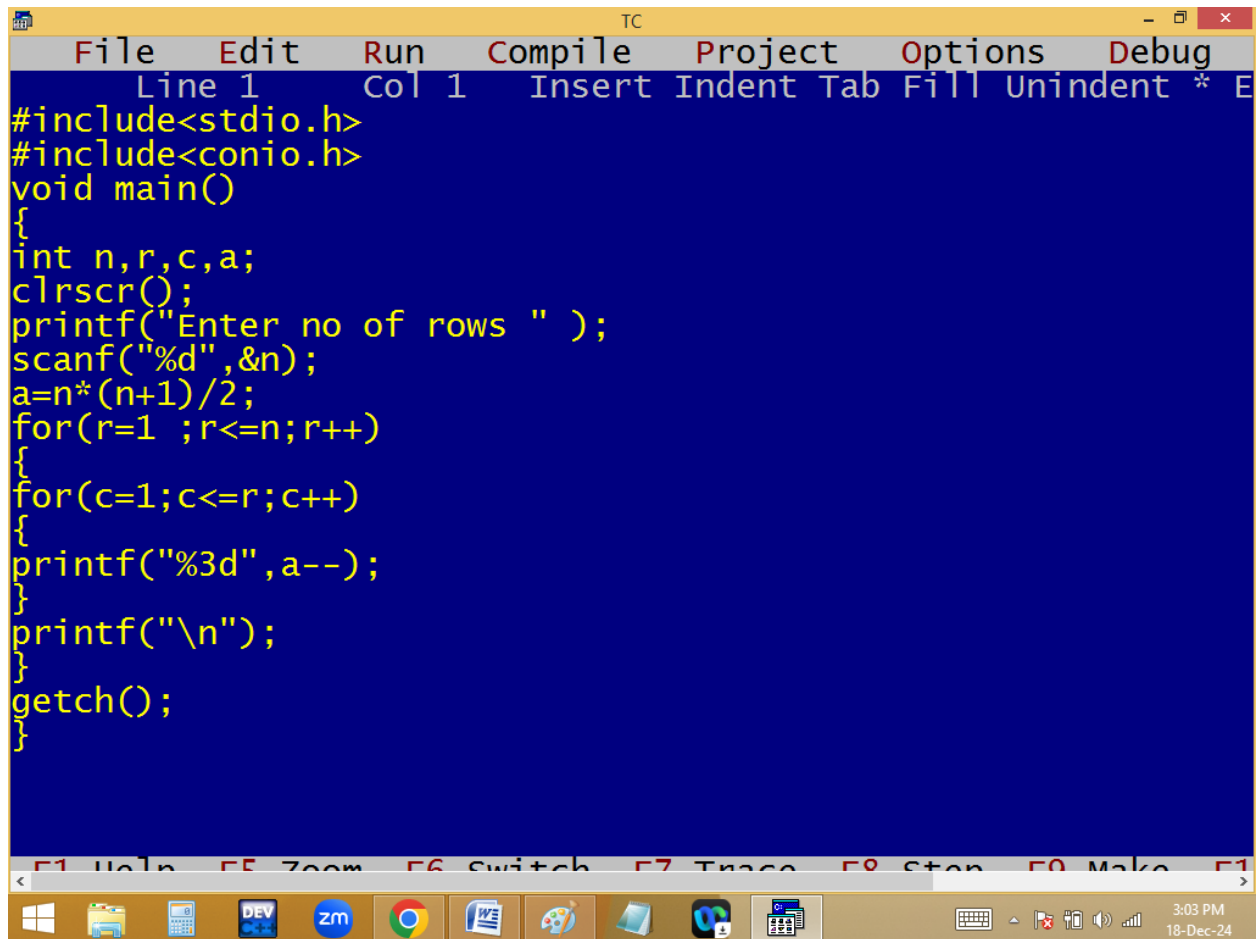
```
TC
File Edit Run Compile Project Options Debug
Line 1 Col 3 Insert Indent Tab Fill Unindent * E
/* Floyd's Triangle */
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c,a=1;
clrscr();
printf("Enter no of rows " );
scanf("%d",&n);
for(r=1 ;r<=n;r++)
{
for(c=1;c<=r;c++)
{
printf("%3d",a++);
}
printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

2:53 PM
18-Dec-24

```
TC
Enter no of rows 4
1
2 3
4 5 6
7 8 9 10
```



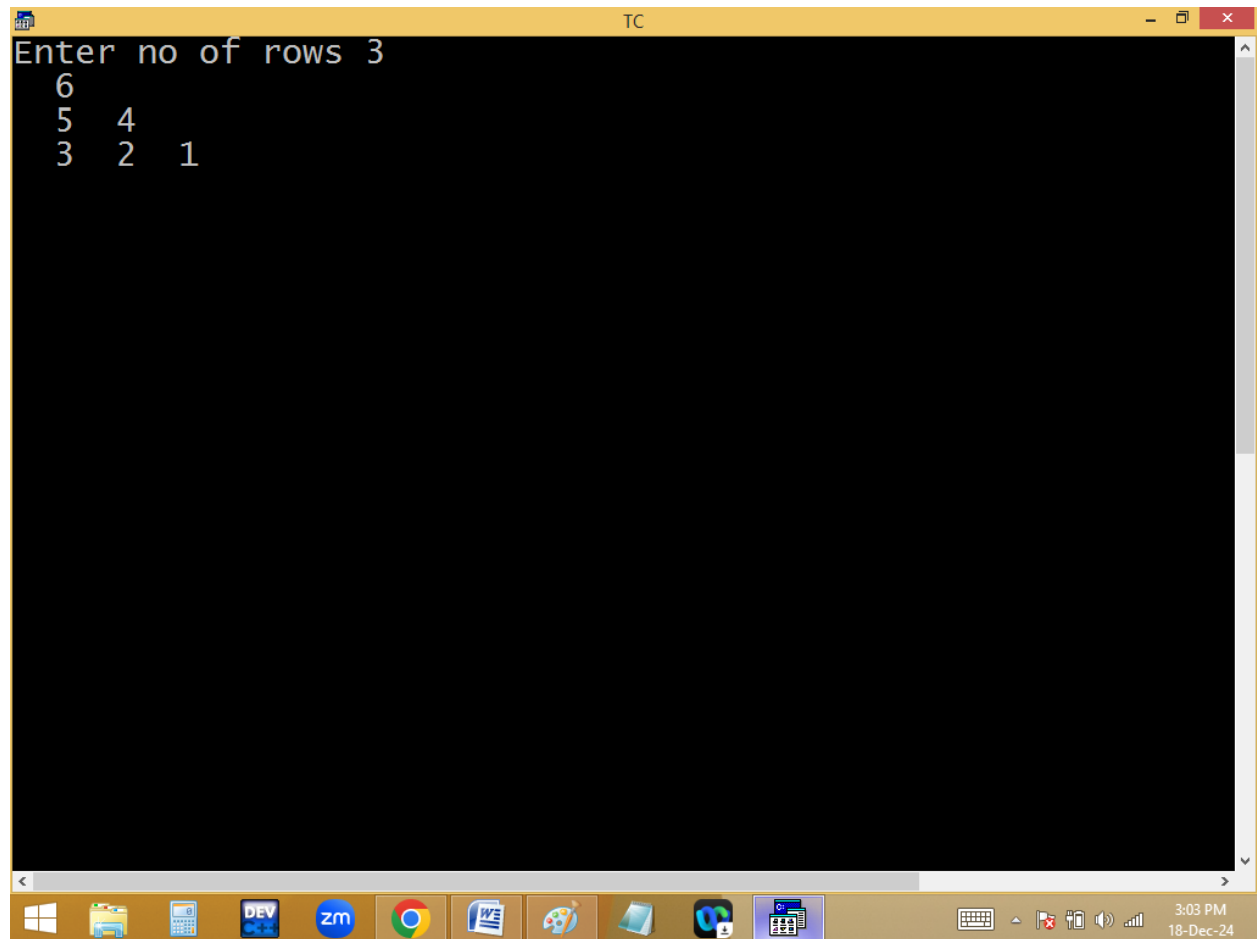


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar is a status bar showing "Line 1", "Col 1", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

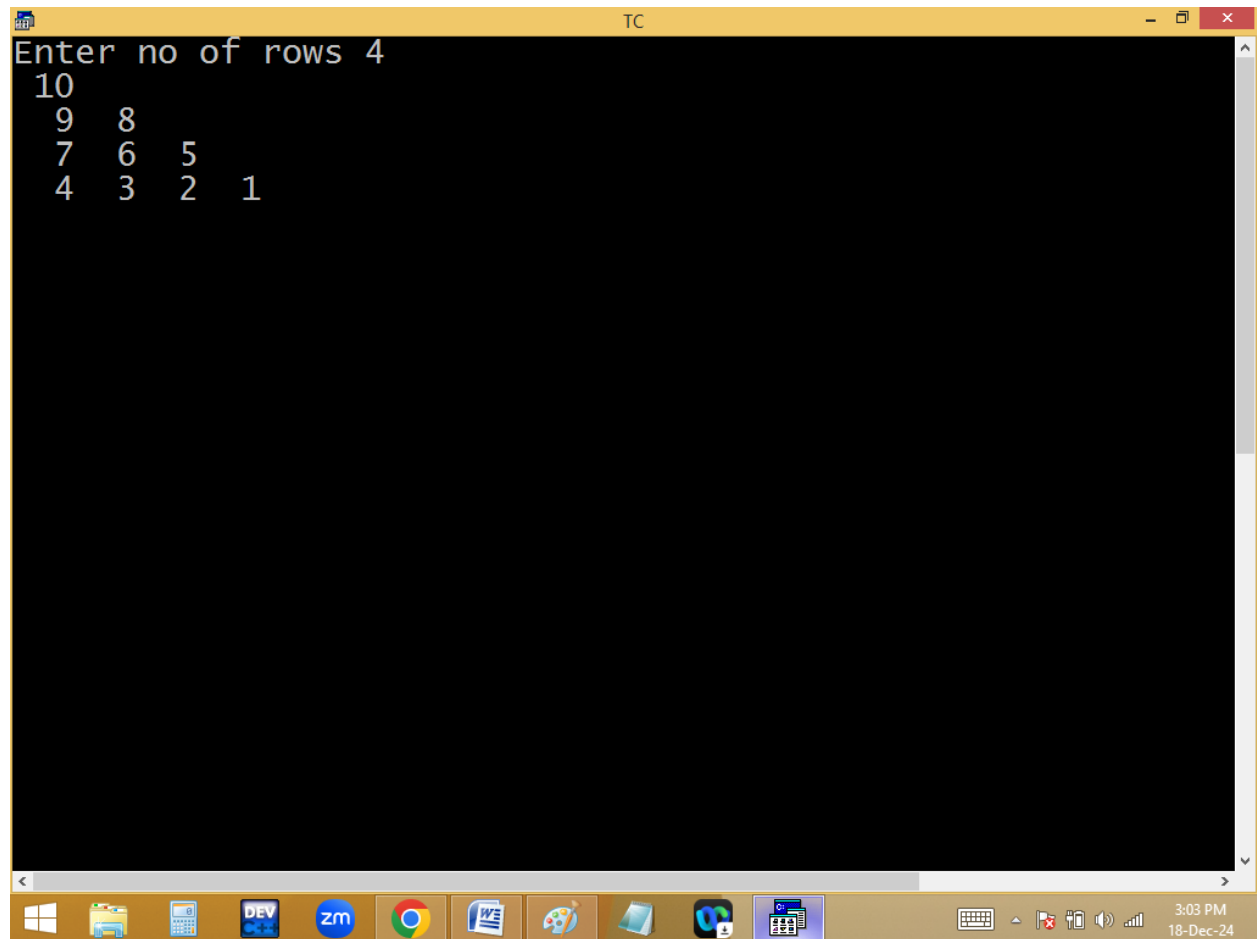
```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c,a;
clrscr();
printf("Enter no of rows " );
scanf("%d",&n);
a=n*(n+1)/2;
for(r=1 ;r<=n;r++)
{
for(c=1;c<=r;c++)
{
printf("%3d",a--);
}
printf("\n");
}
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "3:03 PM" and the date "18-Dec-24".

```
TC
Enter no of rows 3
6
5 4
3 2 1
```



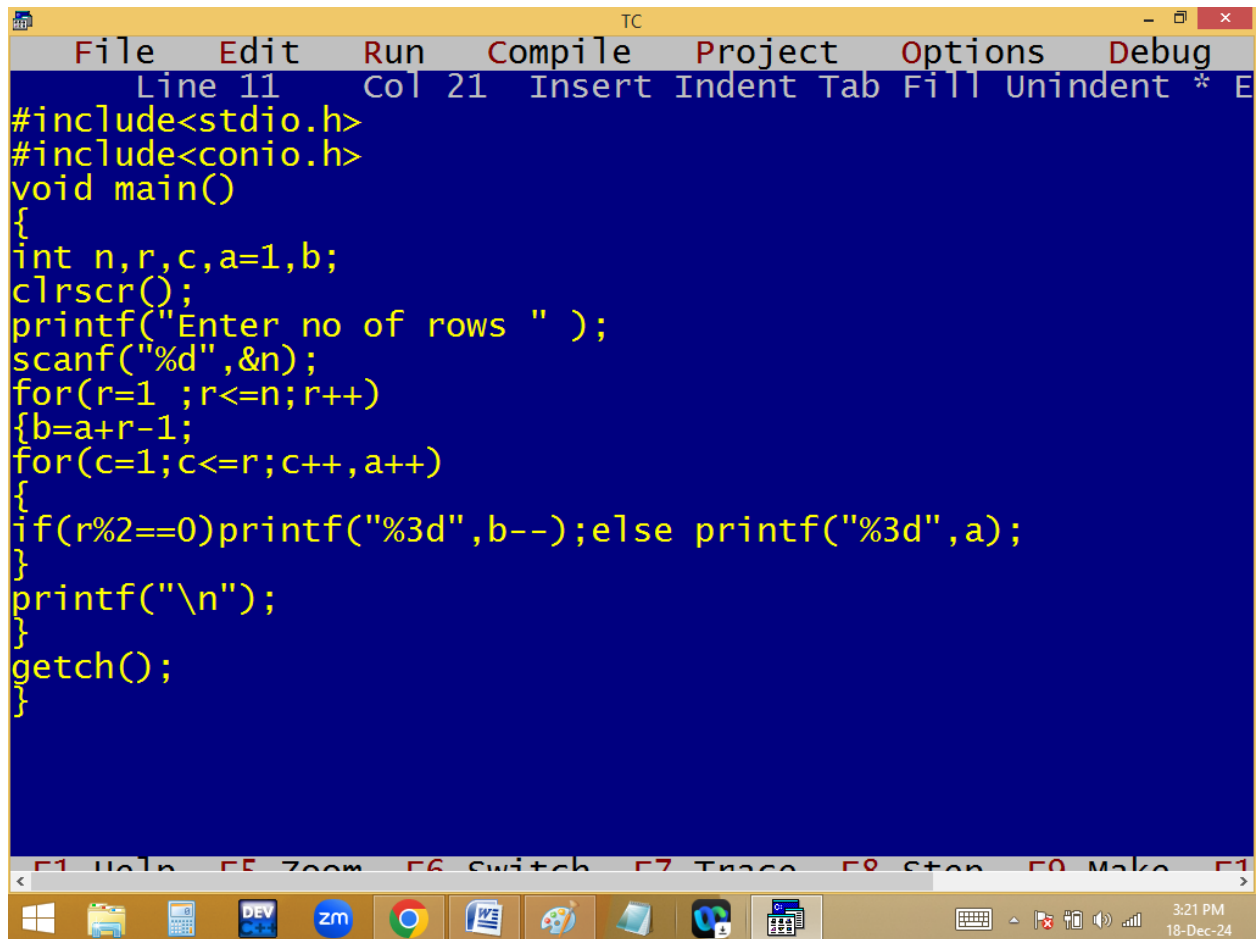

```
TC
Enter no of rows 4
10
9 8
7 6 5
4 3 2 1
```



The image shows a Windows 10 desktop environment. A terminal window titled 'TC' is open, displaying a Pascal's triangle pattern. The pattern consists of four rows of numbers, with the first row containing '10' and the subsequent rows containing pairs of numbers, then triplets, and finally a single number. The desktop taskbar at the bottom includes icons for the Start menu, File Explorer, Calculator, DEV, ZM, Chrome, Word, Paint, Edge, and a calendar. The system tray on the right shows the time as 3:03 PM and the date as 18-Dec-24.

$a = n * (n + 1) / 2;$ $n = 3$
 $3 * 4 / 2 = 6$ $\rightarrow$ 6
5 4
 $a--;$ 3 2 1

$4 * 5 / 2 = 10$ $\rightarrow$ $n = 4$ 10
9 8
7 6 5
4 3 2 1

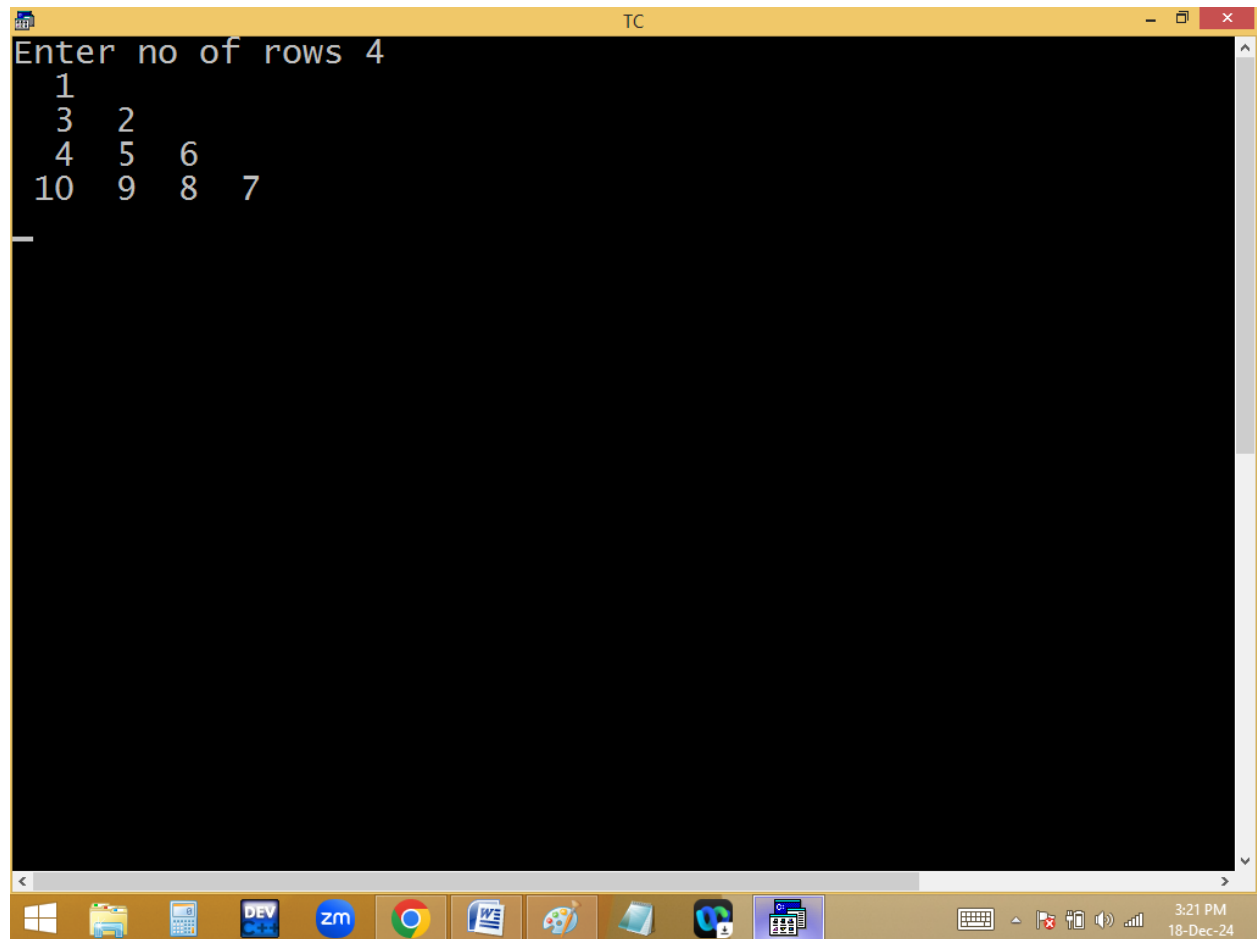


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 11", "Col 21", and a list of editing commands: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main text area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c,a=1,b;
clrscr();
printf("Enter no of rows " );
scanf("%d",&n);
for(r=1 ;r<=n;r++)
{b=a+r-1;
for(c=1;c<=r;c++,a++)
{
if(r%2==0)printf("%3d",b--);else printf("%3d",a);
}
printf("\n");
}
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions and a status bar. The status bar on the right shows the time "3:21 PM" and the date "18-Dec-24".

```
TC
Enter no of rows 4
1
3 2
4 5 6
10 9 8 7
```



The image shows a Windows 10 desktop environment. A terminal window titled "TC" is open, displaying a pattern of numbers. The pattern consists of four rows of numbers, where the first row has one number (1), the second row has two numbers (3, 2), the third row has three numbers (4, 5, 6), and the fourth row has four numbers (10, 9, 8, 7). The taskbar at the bottom of the screen contains several application icons: Windows, File Explorer, Calculator, DEV, Zoom, Chrome, Word, Paint, Edge, and a calendar. The system clock in the bottom right corner indicates the time is 3:21 PM on 18-Dec-24.

```

Enter no of rows 10
1
3 2
4 5 6
10 9 8 7
11 12 13 14 15
21 20 19 18 17 16
22 23 24 25 26 27 28
36 35 34 33 32 31 30 29
37 38 39 40 41 42 43 44 45
55 54 53 52 51 50 49 48 47 46

```

```

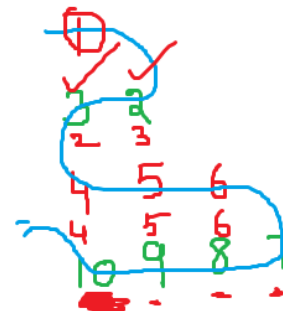
for ( r=1; r<=4; r++)
{
    7+4-1=10
    b = a+r-1;
    a+a-1=3 a++
    for( c=1;c<=r;c++)
    {
        if(r%2==0)p(b--); ✓
        else p(a);
    }
}

```

Y
1

Q
1 2

7

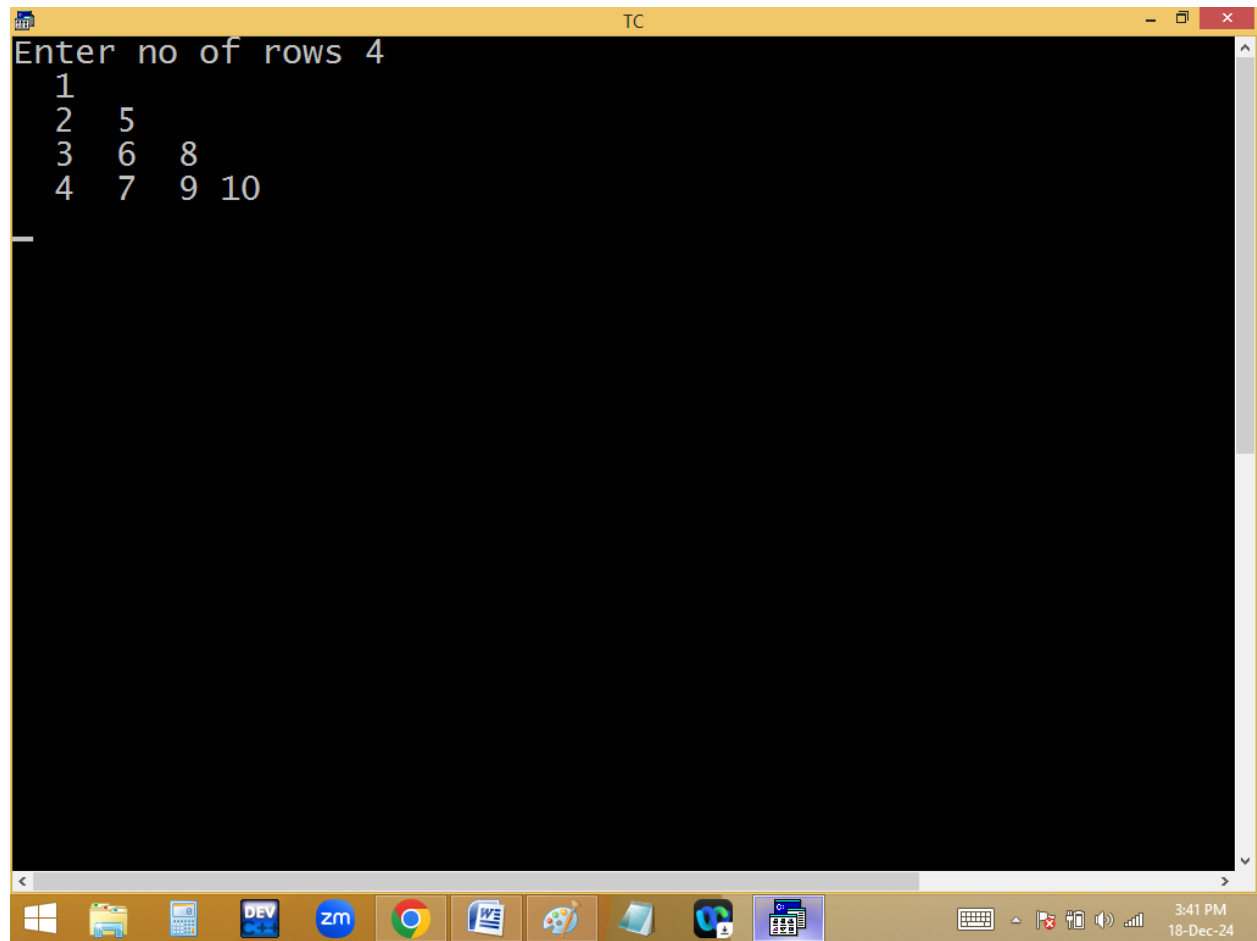


```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 17 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c,a=1,b;
clrscr();
printf("Enter no of rows " );
scanf("%d",&n);
for(r=1 ;r<=n;r++)
{ a=n-1; b=a+r;
for(c=1;c<=r;c++)
{
if(c==1)printf("%3d",r);
else {printf("%3d",b);a--;b=a+b;}
}
printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

3:41 PM
18-Dec-24

```
TC
Enter no of rows 4
1
2 5
3 6 8
4 7 9 10
```



```
TC
Enter no of rows 10
1
2 11
3 12 20
4 13 21 28
5 14 22 29 35
6 15 23 30 36 41
7 16 24 31 37 42 46
8 17 25 32 38 43 47 50
9 18 26 33 39 44 48 51 53
10 19 27 34 40 45 49 52 54 55
```


1
 ✓ 2 5
 3 6 8
 4 7 9 10
 10

$\frac{n}{4}$	$\frac{a=n-1}{3}$	$\frac{b=a+1}{a+3=5}$
	2 ✓	$3+3=6+2=8$
	1	

$$\begin{array}{r}
 3+4=7 \\
 \underline{2} \\
 9 \\
 \underline{1} \\
 10
 \end{array}$$

```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 10 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows " );
scanf("%d",&n);
for(r=1 ;r<=n;r++)
{
for(c=1;c<=r;c++)
{
if(((r+c)%2==0))printf("$ ");
else printf("* ");
}
printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

3:46 PM
18-Dec-24

```
TC
Enter no of rows 10
$
* $
$ * $
* $ * $
$ * $ * $
* $ * $ * $
$ * $ * $ * $
* $ * $ * $ * $
$ * $ * $ * $ * $
* $ * $ * $ * $ * $
$ * $ * $ * $ * $ * $
```

```
if( (r+c)%2==0)p($);
else p(*);
```

\$
1+1=2

* \$
2+1=3 2+2=4

\$ * \$
3+1=4 3+2=5 3+3=6

* \$ * \$

Home work:

1 2 3 4

1 2 3

1 2

1

4 3 2 1

4 3 2

4 3

4

4

34

234

1234

